

* 10/13 : E.M., thermo, QM *

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Plan: QM, classical, thermo

2021 / Q10 $\begin{cases} \psi(x) = A \left[e^{\frac{i8\pi x}{a}} - e^{\frac{i4\pi x}{a}} \right], & |x| \leq \frac{a}{2} \\ \psi(x) = 0, & |x| > \frac{a}{2} \end{cases}$

$\int |\psi|^2 dx = 1$ $|\psi|^2 = \psi^*(x) \psi(x)$

$|\psi|^2$: prob. density $\psi^*(x) = A \left(e^{-\frac{i8\pi x}{a}} - e^{-\frac{i4\pi x}{a}} \right)$

$\psi^*(x) \psi(x) = A^2 \left(e^{-\frac{i8\pi x}{a}} - e^{-\frac{i4\pi x}{a}} \right) \left(e^{\frac{i8\pi x}{a}} - e^{\frac{i4\pi x}{a}} \right)$

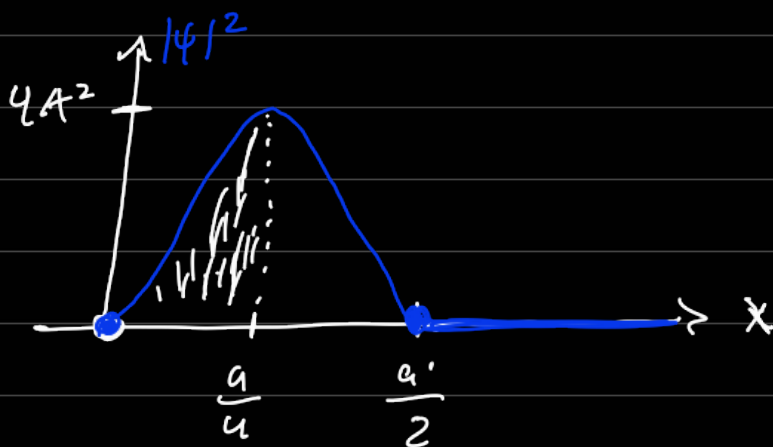
$$= A^2 \left(e^{-\frac{i8\pi x}{a}} e^{\frac{i8\pi x}{a}} - e^{-\frac{i4\pi x}{a}} e^{\frac{i8\pi x}{a}} - e^{-\frac{i8\pi x}{a}} e^{\frac{i4\pi x}{a}} + e^{-\frac{i4\pi x}{a}} e^{\frac{i4\pi x}{a}} \right)$$

$$= A^2 (1 - e^{+i4\pi x/a} - e^{-i4\pi x/a} + 1)$$

$$= A^2 (2 - (e^{i4\pi x/a} + e^{-i4\pi x/a}))$$

$2 \cos(\theta) = (e^{i\theta} + e^{-i\theta}) \quad \theta = \frac{4\pi x}{a}$

$|\psi|^2 = A^2 (2 - 2 \cos(\frac{4\pi x}{a}))$
 $= 2A^2 (1 - \cos(\frac{4\pi x}{a}))$



$|\psi|^2(0) = 2A^2 (1 - 1) = 0$

$\psi(x) : \frac{a}{2}$

$|\psi|^2(\frac{a}{2}) = 2A^2 (1 - \cos(\frac{4\pi}{a}(\frac{a}{2}))) = 0$

$(1 - \cos(2\pi)) = 0$

$$(1 - 1) = 0$$

$$|\psi|^2\left(\frac{a}{4}\right) = 2A^2 \left(1 - \cos\left(\frac{u\pi}{a}\left(\frac{a}{4}\right)\right)\right) = 2A^2(1 - \cos(\pi)) = 2A^2(1 - (-1)) = 4A^2$$

$$P(|x| \leq \frac{a}{4}) = \int_0^{a/4} |\psi|^2 dx \quad \int_0^{a/2} dx - \int_0^{a/2} \cos(x) dx$$

$$P_{\text{tot}} = 1 \stackrel{!}{=} \int_0^{a/2} |\psi|^2 dx = 2A^2 \int_0^{a/2} \left(1 - \cos\left(\frac{u\pi x}{a}\right)\right) dx$$

$$2A^2 = \frac{1}{\int_0^{a/2} \left(1 - \cos\left(\frac{u\pi x}{a}\right)\right) dx}$$

$$\rightarrow 2A^2 \int_0^{a/4} \left(1 - \cos\left(\frac{u\pi x}{a}\right)\right) dx$$

change variables

$$y = 2x$$

$$x = 0$$

$$y = 2\left(\frac{a}{4}\right) = a/2$$

$$dx = \left(\frac{1}{2}\right) dy$$

$$P(|x| \leq \frac{a}{4}) = \frac{\int_0^{a/4} \left(1 - \cos\left(\frac{u\pi x}{a}\right)\right) dx}{\int_0^{a/2} \left(1 - \cos\left(\frac{u\pi x}{a}\right)\right) dx}$$

$$= \frac{\frac{1}{2} \int_0^{a/2} \left(1 - \cos\left(\frac{u\pi y}{a}\right)\right) dy}{\int_0^{a/2} \left(1 - \cos\left(\frac{u\pi y}{a}\right)\right) dy}$$

$$= \frac{1}{2}$$

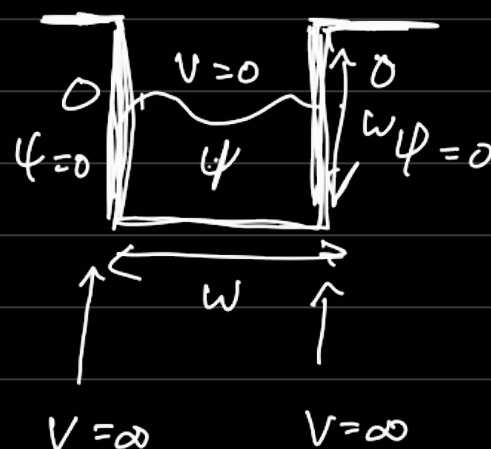
2022]

$$\textcircled{9} \psi(x, y, z)$$

$$\hat{H}\psi = E_n \psi$$

$$\hat{H} = \frac{\vec{p}^2}{2m} + V(x, y, z)$$

$$\frac{1}{2m} \left(\underbrace{p_x^2 + p_y^2 + p_z^2}_{\sum_{i=1}^3 p_i^2} \right)$$



$$\hat{H}\psi = \left(\frac{p^2}{2m} + V\right) \psi = E_n \psi$$

Boundary $V=\infty$:

$$V \rightarrow \infty \quad \left(\frac{1}{V} \frac{p^2}{2m} + 1\right) \psi = \frac{E_n}{V} \psi$$

$$\psi = 0$$

In well: $V=0$

$$\left(\frac{p^2}{2m} + 0\right)\psi = E\psi \quad \text{or}$$

$$\frac{p^2}{2m} \psi = E\psi$$

$$p_x = -i\hbar \frac{\partial}{\partial x} \quad p_y = -i\hbar \frac{\partial}{\partial y} \quad , \quad p_z = -i\hbar \frac{\partial}{\partial z} \quad *$$

$$p_x^2 = i^2 \hbar^2 \frac{\partial^2}{\partial x^2} = -\hbar^2 \frac{\partial^2}{\partial x^2}$$

$$\frac{-\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi = E\psi \quad *$$

$$\text{Sep. sol}^n: * \psi(x, y, z) = X(x) Y(y) Z(z) \quad *$$

$$\text{LHS} \quad -\frac{\hbar^2}{2m} \left(Y(y) Z(z) \left(\frac{\partial^2 X}{\partial x^2} \right) + X(x) Z(z) \left(\frac{\partial^2 Y}{\partial y^2} \right) + X(x) Y(y) \left(\frac{\partial^2 Z}{\partial z^2} \right) \right)$$

$$\text{RHS} \quad E (X Y Z)$$

divide by $(X Y Z)$

$$-\frac{\hbar^2}{2m} \left(\underbrace{\frac{1}{X} \frac{\partial^2 X}{\partial x^2}}_x + \underbrace{\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2}}_y + \underbrace{\frac{1}{Z} \frac{\partial^2 Z}{\partial z^2}}_z \right) = E \quad \text{const}$$

$$-\lambda_x^2 \quad -\lambda_y^2 \quad -\lambda_z^2$$

$$\frac{\hbar^2}{2m} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2) = E$$

$$\frac{\partial^2 X}{\partial x^2} = -\lambda_x^2 X(x) \Rightarrow X(x) = A \cos(\lambda x) + B \sin(\lambda x)$$

$$* \psi(0, y, z) = 0$$

$$X(0) = A = 0$$

$$0, \pi, 2\pi,$$

$$X(x) = B \sin(\lambda x)$$

$$\sin(0) = 0$$

$$\psi(w, y, z) = 0$$

$$n\pi$$

$$X(w) = B \sin(\lambda w) = 0$$

$$\lambda_x \omega = n_x \pi$$

$$\lambda_x = \frac{1}{\omega} n_x \pi$$

$$\lambda_y = \frac{1}{\omega} n_y \pi$$

$$\lambda_z = \frac{1}{\omega} n_z \pi$$

$$\frac{\hbar^2}{2m} \frac{\pi^2}{\omega^2} (n_x^2 + n_y^2 + n_z^2) = E_N$$

$$N \equiv (n_x^2 + n_y^2 + n_z^2)$$

$$\psi(x, y, z) = \sum_{n_x n_y n_z} \underbrace{C_{n_x n_y n_z}}_{\text{Norm const.}} \left(\sin\left(\frac{n_x \pi x}{\omega}\right) \times \sin\left(\frac{n_y \pi y}{\omega}\right) \times \sin\left(\frac{n_z \pi z}{\omega}\right) \right)$$

